Coupled dipole model for coherent perfect absorption

Summary of these notes

The geometry considered in these notes is a one-dimensional lattice of cylinders with periodicity, a, with each cylinder having a radius, b. Each cylinder is considered to have a vacuum interior (i.e. $\varepsilon(r < b) = 1$) and a monolayer coating with conductivity $\sigma(\omega) = \sigma'(\omega) + i\sigma''(\omega)$. We wish to find the input wavefront and geometry required to perfectly absorb light. Various derivations here mirror results reported in the paper Extraordinary optical reflection from sub-wavelength cylinder arrays (https://opg.optica.org/directpdfaccess/87a6fce8-e489-4bee-bbbfeae2114ce881_89577/oe-14-9-3730.pdf?da=1&id=89577&seq=0&mobile=no).

Section 1 below is a review of the Dyadic Green's function. **Section 2** is a discussion of the Green's function in two dimensions. In **Section 3**, we calculate the lattice-summed Green's function in the far field. In **Section 4**, we calculate the dipole-dipole interaction due to each pair of dipoles in the lattice (for which we must use some numerical tricks). This is done for **s-polarization** (electric field along the cylinder's axis). For **p-polarization**, the sum converges quickly, so we do not bother ourselves with numerical tricks. In **Section 5**, we find analytic forms for the imaginary part of the lattice Green's function sum (dipole-dipole interaction) for **p-polarization**. **Section 6** is just a proof of an identity. **Section 7 and Section 8** derive polarizabilities of cylinders in the presence of radiative losses. **Sections 9 and 10** find the CPA condition.

Notational note

Unless otherwise indicated, $n_m, \varepsilon_m \equiv n_m^2$ refer to indices and permittivities of the scatterers under consideration.

Section 1: From scalar to Dyadic Green's function in 3 dimensions

In the Lorenz gauge, the vector potential is given in terms of the current density via the relation,

$$\mathbf{A}(\mathbf{r},\omega) = \mu_0 \int \mathrm{G}(\mathbf{r},\mathbf{r}',\omega)\mathbf{j}(\mathbf{r}',\omega)\mathrm{d}^3\mathbf{r}' \rightarrow \mathbf{E}(\mathbf{r},\omega) = i\omega\mathbf{A}(\mathbf{r},\omega) - \nabla\Phi(\mathbf{r},\omega)$$

From **Equation 2.90** in **Novotny and Hecht**, we also have that

$$\begin{aligned}\nabla\Phi(\mathbf{r},\omega) = -i(c^2/\varepsilon\omega)\nabla(\nabla\cdot\mathbf{A}(\mathbf{r},\omega)) \rightarrow \mathbf{E}(\mathbf{r},\omega) &= i\omega\left[\mathbf{A}(\mathbf{r},\omega) + \frac{c^2}{\varepsilon\omega^2}\nabla(\nabla\cdot\mathbf{A}(\mathbf{r},\omega))\right] \\ &= i\omega\mu_0\int\left[1+\frac{c^2}{\varepsilon\omega^2}\nabla\nabla\right]\cdot\mathrm{G}(\mathbf{r},\mathbf{r}',\omega)\mathbf{j}(\mathbf{r}',\omega)\mathrm{d}^3\mathbf{r}',\end{aligned}$$

from which we may read off the Dyadic Green's function as (note that all derivatives are with respect to the observation point, $\mathbf{r}$ and not the dipole location, $\mathbf{r}'$):

$$\mathbf{G}(\mathbf{r},\mathbf{r}',\omega) = \left[1+\frac{c^2}{\varepsilon\omega^2}\nabla\nabla\right]\mathrm{G}(\mathbf{r},\mathbf{r}',\omega)$$

Note that the field from a dipole in vacuum is given by the quantity:

$$\mathbf{E}_{\text{dipole}}(\mathbf{r}) = \mu_0\omega^2\left[1+\frac{c^2}{\varepsilon\omega^2}\nabla\nabla\right]\mathrm{G}(\mathbf{r},\mathbf{r}')\mathbf{p}(\mathbf{r}') = \frac{1}{\varepsilon_0}\left[\frac{\omega^2}{c^2}+\nabla\nabla\right]\mathrm{G}(\mathbf{r},\mathbf{r}')\mathbf{p}(\mathbf{r}')$$

Note that $\mathbf{p}(\mathbf{r}')$ has units of $\varepsilon_0 \times \text{Electric field} \times \text{Volume.}$

We also note that, although the scalar Green's function is usually written as $e^{i\omega r/c}/(4\pi r)$, we may write this also in terms of the spherical Hankel functions by noting that:

$$\frac{e^{i\omega r/c}}{4\pi r} = \left[\frac{\cos(\omega r/c) + i\sin(\omega r/c)}{\omega r/c}\right]\frac{\omega}{4\pi c} = -y_0(\omega r/c) + ij_0(\omega r/c) = \frac{ik}{4\pi}h_0^{(1)}(r\omega/c)$$

Section 2: From scalar to Dyadic Green's function in 2 dimensions.

The Green's function to the scalar Helmholz equation in two dimensions is given by (**Page 1362 in Morse and Feshbach**):

$$\mathrm{G}(\mathbf{r},\mathbf{r}') = \frac{i}{4}H_0^{(1)}(|\mathbf{r}-\mathbf{r}'|\omega/c)$$

Section 3: Far-field expansion of lattice sum in two dimensions

In this section, we wish to calculate the field radiated from a one-dimensional lattice of dipole cylinders (cylinder axis along the $\mathbf{e}_z$ direction and periodicity direction along the $\mathbf{e}_x$ axis). In particular, we are concerned with the far-field ($|y| >> \lambda$, with $\lambda = 2\pi c/\omega$ denoting the wavelength of light under consideration).

We start with the expression after **Equation 11.25** on **Page 1362** of **Morse an Feshbach** (**Part 2 of the two-part series**) for the scalar Green's function in two dimensions, written out in Cartesian Fourier components:

$$\sum_n \mathrm{G}(\mathbf{r}, \mathbf{r}_n) = \frac{1}{4\pi^2} \sum_n \int\int \frac{e^{i\mathbf{k}\cdot(\mathbf{r}-\mathbf{r}_n)}}{|\mathbf{k}|^2 - (\omega/c)^2} \mathrm{d}k_x \mathrm{d}k_y,$$

where $\mathbf{r} - \mathbf{r}_n = (x - na, y)$, with $\mathbf{r}_n$ labeling a lattice point at coordinate $(x = na, y = 0)$ and $\mathbf{k} = (k_x, k_y)$. We note that this Fourier decomposition for the Green's function may be trivially derived:

$$\nabla^2 \mathrm{G}(\mathbf{r}, \mathbf{r}_0) + (\omega/c)^2 \mathrm{G}(\mathbf{r}, \mathbf{r}_0) = -\delta(\mathbf{r} - \mathbf{r}_0) \rightarrow \left[|\mathbf{k}|^2 - (\omega/c)^2\right] \mathrm{G}(\mathbf{k}) = 1,$$
$$\text{where } \mathrm{G}(\mathbf{r}, \mathbf{r}_0) = \frac{1}{4\pi^2} \int e^{i\mathbf{k}\cdot(\mathbf{r}-\mathbf{r}_0)} \mathrm{G}(\mathbf{k}) \mathrm{d}^2\mathbf{k}$$

We first perform the sum over the real space lattice points, (of which there are N total),

$$\sum_{n=0}^{N-1} e^{i(x-na)k_x} = N \sum e^{ik_x x} \delta_{k_x, 2\pi m/a} = \frac{2\pi N}{L} \sum_m e^{ik_x x} \delta(k_x - 2\pi m/a),$$

where a denotes the lattice spacing and we used the following relation between the Kronecker delta and the delta function in reciprocal space:

$$(L/2\pi) = (L/2\pi) \int \delta(k - k') \mathrm{dk} = \sum_k \delta(k - k') \rightarrow \delta_{k,k'} = (2\pi/L)\delta(k - k')$$

Therefore, our lattice=summed scalar Green's function is given by:

$$\frac{1}{2\pi a}\int_{-\infty}^{\infty} e^{ik_y y}\sum_m e^{i2\pi m x/a}\frac{1}{k_y^2+(2\pi m/a)^2-(\omega/c)^2}\mathrm{d}k_z$$

At this point, we expand denominator into a sum of simple poles:

$$\frac{1}{x^2-a^2}=\frac{1}{2a}\left[\frac{1}{x-a}-\frac{1}{x+a}\right]\rightarrow\frac{1}{k_y^2+(2\pi m/a)^2-(\omega/c)^2}=$$
$$\frac{1}{2\sqrt{(\omega/c)^2-(2\pi m/a)^2}}\left[\frac{1}{k_y-\sqrt{(\omega/c)^2-(2\pi m/a)^2}}-\frac{1}{k_y+\sqrt{(\omega/c)^2-(2\pi m/a)^2}}\right],$$

meaning our lattice-summed scalar Green's function is given by:

$$\frac{1}{4\pi a}\sum_m\frac{e^{i2\pi m x/a}}{\sqrt{(\omega/c)^2-(2\pi m/a)^2}}\int_{-\infty}^{\infty}e^{ik_y y}\left[\frac{1}{k_y-\sqrt{(\omega/c)^2-(2\pi m/a)^2}}-\frac{1}{k_y+\sqrt{(\omega/c)^2-(2\pi m/a)^2}}\right]\mathrm{d}k_y$$

We focus on the term corresponding to lowest order diffraction, i.e. $m=0$. In order to evaluate the contour integral, we need to add a small imaginary part to the angular frequency, $\omega\rightarrow\omega+i\delta$. This corresponds to a Green's function having no poles in the upper half of the complex plane (i.e. being causal). For $y>0$, we must close the contour in the upper half of the complex plane, which results in picking up a residue at $k_y=(\omega+i\delta)/c$. For $y<0$, we close the contour in the lower half of the complex plane and pick up a residue at $k_z=-(\omega+i\delta)/c$. Our lattice sum then gives:

$$\sum_n \mathrm{G}(\mathbf{r},\mathbf{r}_n)=\frac{i}{2(a\omega/c)}e^{i|y|\omega/c}+\text{terms corresponding to } m\neq 0$$

Note that for $y>0$, this goes as $e^{i2\pi y/\lambda}$ (plane wave propagating in the possitive y direction. And for $y<0$, it corresponds to a plane wave propagating in the minus y direction. Now we consider the terms with $m\neq 0$. If $2\pi/a>\omega/c\leftrightarrow\lambda>a$, we have necessarily that all the poles will lie on the imaginary axis. In particular, for each such term, we will obtain a contribution to the lattice sum given by:

$$\frac{ie^{i2\pi mx/a}}{2a\sqrt{(\omega/c)^2-(2\pi m/a)^2}}e^{i\sqrt{(\omega/c)^2-(2\pi m/a)^2}|y|}$$

However, note that all of these terms are evanescent and do not contribute in the far-field (and we therefore do not have to consider these contributions to the reflection/transmission coefficients).

Section 4: Dipole-dipole lattice sum in two dimensions (s-polarization)

In the previous section, we calculated the field scattered by all lattice sites in the far-field. However, in order to formulate a coupled dipole model, we must calculate the field at each given lattice site due to all other lattice sites. Concretely, we wish to calculate the quantity:

$$\sum_{m\neq n}\mathrm{G}(\mathbf{r}_n,\mathbf{r}_m)=\frac{1}{4\pi^2}\sum_{m\neq .n}\int\int\frac{e^{i\mathbf{k}\cdot(\mathbf{r}_n-\mathbf{r}_m)}}{|\mathbf{k}|^2-(\omega/c)^2}\mathrm{d}k_x\mathrm{d}k_y,$$

We write this as the difference of two terms (defining $k\equiv|\mathbf{k}|$):

$$\sum_{m\neq n}\mathrm{G}(\mathbf{r}_n,\mathbf{r}_m)=-\mathrm{G}(\mathbf{r}_n,\mathbf{r}_n)+\sum_{m}\mathrm{G}(\mathbf{r}_n,\mathbf{r}_m)$$

Evaluation of the first term

The first term, which corresponds to the self energy of the lattice site with itself, may be evaluated as:

$$\lim_{\mathbf{r}\to\mathbf{r}_n}\frac{i}{4}H_0(|\mathbf{r}-\mathbf{r}_n|\omega/c)\approx\frac{i}{4}\left[1+\frac{2i}{\pi}\ln\left(|\mathbf{r}-\mathbf{r}_n|\omega/2c\right)+\frac{2i}{\pi}\gamma_e\right]=$$
$$-\frac{\gamma_e}{2\pi}-\frac{1}{2\pi}\ln\left(|\mathbf{r}-\mathbf{r}_n|\omega/2c\right)+\frac{i}{4},$$

where we used the asymptotic forms of the cylindrical Bessel functions, reported here: https://en.wikipedia.org/wiki/Bessel_function#Hankel_functions (https://en.wikipedia.org/wiki/Bessel_function#Hankel_functions) as well as the relation $H_0=J_0+iY_0$. Note also that γ_e above is the Euler-Macaroni constant (I refuse to look up the spelling of the second guy's name). To tame the logarithmic divergence, we , at this point, write $\ln\left(|\mathbf{r}-\mathbf{r}_n|\omega/2c\right) = \ln(|\mathbf{r}-\mathbf{r}_n|2\pi/a) + \ln(a\omega/4\pi c)$, and, furthermore, for $|\mathbf{r}-\mathbf{r}_n| << 1$, $\ln(2\pi|\mathbf{r}-\mathbf{r}_n|/a|) \approx \ln\left[1-e^{-2\pi|\mathbf{r}-\mathbf{r}_n|/a}\right] = -\sum_{m>0} e^{-2\pi|\mathbf{r}-\mathbf{r}_n|m/a}/m$.

Therefore, we write out the entirety of the first term as:

$$-\frac{\gamma_e}{2\pi} + \frac{i}{4} - \frac{1}{2\pi}\ln(a\omega/4\pi c) + \frac{1}{a}\sum_{m>0}\frac{e^{-2\pi|\mathbf{r}-\mathbf{r}_n|m/a}}{2\pi m/a}$$

Evaluation of the second term

The second term corresponds to all lattice points, including $m = n$:

$$\frac{1}{2\pi a}\sum_m \int_{-\infty}^{\infty} \frac{1}{k_y^2 + (2\pi m/a)^2 - (\omega/c)^2}\mathrm{d}k_y$$

We now note that $\int \frac{1}{x^2+a^2} = \left[\tan^{-1}(\infty) - \tan^{-1}(-\infty)\right]/a = \pi/a$

By assumption, we had that $2\pi|m|/a > \omega/c$ for all $|m| > 0$. Therefore the second term may be written as:

$$\sum_m \frac{1}{2a\sqrt{(2\pi m/a)^2 - (\omega/c)^2}} + \frac{i}{2(a\omega/c)}$$

Combination of the first and second terms

Therefore, all terms combined give us:

$$\sum_{m\neq n} \mathrm{G}(\mathbf{r}_n, \mathbf{r}_m) = \frac{i}{2}\left[\frac{1}{a\omega/c} - \frac{1}{4}\right] + \frac{1}{a}\sum_{m=1}^{\infty}\left[\frac{1}{\sqrt{(2\pi m/a)^2 - (\omega/c)^2}} - \frac{1}{2\pi m/a}\right] + \frac{1}{2\pi}\left[\gamma_e + \ln(a\omega/4\pi c)\right],$$

which is exactly **Equation 7** in Extraordinary optical reflection from sub-wavelength cylinder arrays (https://opg.optica.org/directpdfaccess/87a6fce8-e489-4bee-bbbfeae2114ce881_89577/oe-14-9-3730.pdf?da=1&id=89577&seq=0&mobile=no).

For future reference, we also write the above in terms of the wavelength:

$$\sum_{m\neq n} \mathrm{G}(\mathbf{r}_n, \mathbf{r}_m) = \frac{i}{2}\left[\frac{(\lambda/a)}{2\pi} - \frac{1}{4}\right] + \frac{1}{2\pi}\sum_{m=1}^{\infty}\left[\frac{1}{\sqrt{m^2-(a/\lambda)^2}} - \frac{1}{m}\right] + \frac{1}{2\pi}\left[\gamma_e + \ln(a/2\lambda)\right]$$

Section 5: Closed form solution of imaginary part of lattice Green's function sum

In this section, we wish to find an analytic form for the Green's function corresponding to **p-polarization**. Explicitly, we wish to calculate the sum of the quantity $\partial_y^2 \Im \mathrm{G}(\mathbf{r}_n, \mathbf{r}_m)$ over all $m \neq n$.
We first calculate the imaginary part of the Green's function corresponding to the self-energy:

$$\lim_{\mathbf{r}\to\mathbf{r}_0} \partial_y^2 \Im \mathrm{G}(\mathbf{r}, \mathbf{r}_0) = \frac{1}{4}\partial_y^2 J_0(k|\mathbf{r}-\mathbf{r}_0|) = \frac{1}{4}\left[\frac{kJ_0'(kr)}{r}(1-(y/r)^2) + k^2(y/r)^2 J_0''(kr)\right] \approx -\frac{1}{8}k^2,$$

where we used $J_0'(z) = -J_1(z) \approx -z/2 + O(z^3)$, which allowed us to write $k(kJ''(kr) - J_0'(kr)/r)(y/r)^2 \approx k^2 O((kr)^2)(y/r)^2$, which vanishes in the limit $r \to 0$ (regardless of how we take the limit). Therefore, the lattice sum's imaginary part will be:

$$-\lim_{\mathbf{r}\to\mathbf{r}_0} \partial_y^2 \Im \mathrm{G}(\mathbf{r}, \mathbf{r}_0) + \sum_m \partial_y^2 \Im \mathrm{G}(\mathbf{r}_n, \mathbf{r}_m) = -k/(2a) + k^2/8$$

The first term in the above, $-k/2a$ corresponds to the zeroth-order diffraction term in the lattice Green's function (which we already derived in a **Section 3**).

Section 6: Derivation of Equation 16

The scattered magnetic field due to a dipole is given by

$$-i\omega\mu_0 \nabla \times \mathrm{G}(\mathbf{r}, \mathbf{r}_0)\mathbf{p}$$

Note that this is entirely consistent with the Green's function for the electric field. In particular, if we wanted to find the electric field corresponding to the magnetic field above, we would calculate:

$$\mathbf{E} = \frac{1}{\varepsilon_0}\nabla\times\nabla\times \mathrm{G}(\mathbf{r},\mathbf{r}_0)\mathbf{p} = \mu_0 c^2\left[\frac{\omega^2}{c^2}\mathrm{G}(\mathbf{r},\mathbf{r}_0)\mathbf{p} + \nabla\nabla\cdot\mathrm{G}(\mathbf{r},\mathbf{r}_0)\mathbf{p}\right]$$

For an incident magnetic field polarized in the e_z direction, we have that $\mathbf{p} = \alpha\cdot\mathbf{E} = \frac{i}{\omega\varepsilon_0}(\varepsilon_0\alpha_{xx}\partial_y H_z, -\varepsilon_0\alpha_{yy}\partial_x H_z, 0)$

From this, we have that the scattered magnetic field is given by:

$$\mathbf{H}_{\mathrm{scat},i} = \varepsilon_{ijk}\partial_j\mathrm{G}(\mathbf{r},\mathbf{r}_0)\mathbf{p}_k \to \mathbf{H}_{\mathrm{scat},z} = -\partial_x\mathrm{G}(\mathbf{r},\mathbf{r}_0)\alpha_{yy}\partial_x H_z - \partial_y\mathrm{G}(\mathbf{r},\mathbf{r}_0)\alpha_{xx}\partial_y H_z,$$

which is **Equation 16** in Extraordinary optical reflection from sub-wavelength cylinder arrays (https://opg.optica.org/directpdfaccess/87a6fce8-e489-4bee-bbbfeae2114ce881_89577/oe-14-9-3730.pdf?da=1&id=89577&seq=0&mobile=no).

Section 7: Retarded polarizability for a cylinder

In this section, we wish to calculate the retarded polarizability of a cylinder (perpendicular to the cylinder axis) in terms of the cylindrical Mie coefficients. We do this as follows: First, we calculate the field that would be scattered by a pure electric dipole, with polarizability α in the presence of an incident magnetic field polarized along the cylinder's axis, $\mathbf{H} = H_{\mathrm{inc,\ z}}\mathbf{e}_z$. We then match this to the scattered field that is predicted by the Mie coefficients.
Concretely, we consider an electric dipole, $\mathbf{p} = \varepsilon_0\alpha_{xx}E_{inc,x}\mathbf{e}_x + \varepsilon_0\alpha_{yy}E_{inc,y}\mathbf{e}_y$, where, as in the previous sections, $\alpha_{xx} = \alpha_{yy} \equiv \alpha$. Furthermore, $\mathbf{E}_{\mathrm{inc}} = (i/\omega\varepsilon_0)(\partial_y H_{\mathrm{inc},z}, -\partial_x H_{\mathrm{inc},z})$ (and the derivatives are understood to be evaluated at the coordinate origin. The current density of such a dipole is given by $\mathbf{J} = -i\omega\mathbf{p}\delta(\mathbf{r}-\mathbf{r}_0) = \alpha(\partial_y H_{\mathrm{inc},z}, -\partial_x H_{\mathrm{inc},z})$, where $\mathbf{r}_0$ denotes the location of the dipole. At this point, we may find the scattered magnetic field, which is also polarized along the cylinder's axis: $H_{\mathrm{scat},z} = (\partial_x G(\mathbf{r},\mathbf{r}_0)J_y(\mathbf{r}_0), -\partial_y G(\mathbf{r},\mathbf{r}_0)J_x(\mathbf{r}_0))$ (where it's

understood that the derivatives are with respect to the observation point, $\mathbf{r}$ and that (and $G(\mathbf{r}, \mathbf{r}_0)$ is the scalar Green's function). Thus, we may write the scattered field in terms of the incident field as:

$$\mathbf{H}_{\text{scat}}(\mathbf{r}) = -\alpha\Big[\partial_x \mathrm{G}(\mathbf{r}, \mathbf{r}_0)\partial_x H_{\text{inc},z}(\mathbf{r}_0) + \partial_y \mathrm{G}(\mathbf{r}, \mathbf{r}_0)\partial_y H_{\text{inc},z}(\mathbf{r}_0)\Big]\mathbf{e}_z = \\ \frac{i\alpha k}{4} H_1^{(1)}(k|\mathbf{r}-\mathbf{r}_0|)\Bigg[\cos(\theta)\partial_x H_{\text{inc},z}(\mathbf{r}_0) + \sin(\theta)\partial_y H_{\text{inc},z}(\mathbf{r}_0)\Bigg]\mathbf{e}_z$$

In order to avail ourselves of the Mie coefficients, we take $H_{\text{inc},z}(\mathbf{r}) = 2J_1(k|\mathbf{r}-\mathbf{r}_0|)e^{i\theta}$ as the incident magnetic field. In addition, in the above we introduced the angle, θ, which is given by $\tan(\theta) = (y-y_0)/(x-x_0)$. In order to calculate the derivatives, we use the chain rule to write $\partial_x = \cos(\theta)\partial_r - \frac{\sin(\theta)}{r}\partial_\theta$ and $\partial_y = \sin(\theta)\partial_r + \frac{\cos(\theta)}{r}\partial_\theta$, from which we immediately obtain:

$$s - 1 = \frac{i\alpha k^2}{2}\Bigg[J_1'(0)\Bigg] \to \frac{n_m J_1(n_m kb)J_l'(kb) - J_1'(n_m kb)J_1(kb)}{n_m J_1(n_m kb)H_1^{(1)}(kb)' - J_1'(n_m kb)H_1^{(1)}(kb)} = \\ -\frac{i\alpha k^2}{4}\Bigg[J_1'(0)\Bigg],$$

where we have introduced the Mie scattering coefficient, s (which is the coefficient of the outgoing Hankel function in the presence of an incoming Hankel function. i.e. the field outside the cylinder is given by $H_1^{(2)}(k|\mathbf{r}-\mathbf{r}_0|) + sH_1^{(1)}(k|\mathbf{r}-\mathbf{r}_0|)$. In addition, we have set the index of refraction outside the cylinder to unity (corresponding to vacuum) and the index of the cylinder to n_m. In order to sanity check our results, we examine the small cylinder, quasistatic, limit. In this limit, we may set $H_1 \approx iY_1, H_2 \approx -iY_1$, giving us the following expression:

$$\frac{n_m J_1(n_m kb)J_1'(kb) - J_1'(n_m kb)J_1(kb)}{n_m J_1(n_m kb)Y_1'(kb) - J_1'(n_m kb)Y_1(kb)} \approx \frac{\alpha k^2}{4}J_1'(0)$$

We use the expansions, $J_1(z) \approx z/2, Y_1(z) \approx -\frac{2}{\pi z}$, from which we obtain:

$$\frac{n_m^2(kb)/4-(kb)/4}{n_m^2/(\pi kb)+1/(\pi kb)}=\frac{\alpha k^2}{8}\to\alpha=(2\pi b^2)\frac{n_m^2-1}{n_m^2+1}=(2\pi b^2)\frac{\varepsilon_m-1}{\varepsilon_m+1},$$

where $\varepsilon_m=n_m^2$ is the relative permittivity of the cylinder.
We may also introduce the effect of radiative loss, which gives us:

$$\frac{n_m^2(kb)/4-(kb)/4}{n_m^2/(\pi kb)+1/(\pi kb)-in_m^2(kb)/4+i(kb)/4}=\frac{\alpha k^2}{8}\to$$
$$\alpha=\frac{(\varepsilon_m-1)(2\pi b^2)}{(\varepsilon_m+1)-i(\varepsilon_m-1)\pi(kb)^2/4}$$

This result is the same as **Equation 17** in the paper Extraordinary optical reflection from sub-wavelength cylinder arrays (https://opg.optica.org/directpdfaccess/87a6fce8-e489-4bee-bbbfeae2114ce881_89577/oe-14-9-3730.pdf?da=1&id=89577&seq=0&mobile=no). Therefore, at this point, we may, with some confidence, write the general formula for a coated cylinder's polarizability (see https://hackmd.io/@aligho/B1Gs5Cz2el#With-a-graphene-cloak-HH_z (https://hackmd.io/@aligho/B1Gs5Cz2el#With-a-graphene-cloak-HH_z) for the generalized Mie scattering coefficient):

$$\alpha=-\frac{4i}{k^2}\left[\frac{n_mJ_1(n_mkb)H_1^{(2)}(kb)'-H_1^{(2)}(kb)J_1'(n_mkb)-\frac{i}{Z_sc\varepsilon_0}J_1'(n_mkb)H_1^{(2)}(kb)'}{H_1^{(1)}(kb)J_1'(n_mkb)-n_mH_1^{(1)}(kb)'J_1(n_mkb)+\frac{i}{Z_sc\varepsilon_0}J_1'(n_mkb)H_1^{(1)}(kb)'}-1\right]$$

Section 8: Retarded polarizability along cylinder axis

In this case, the scattered electric field is given by $E_{\text{scat}}=k^2G_0(\mathbf{r},\mathbf{r}_0)\alpha_{zz}E_{\text{inc}}$, where $E_{\text{inc}}=\lim_{r\to 0}2J_0(kr)=2$ and $E_{\text{scat}}=(s-1)H_0^{(1)}(k|\mathbf{r}-\mathbf{r}_0|)$. From this, we obtain:

$$s-1=ik^2\alpha_{zz}/2\to\frac{J_0(n_mkb)J_0(kb)'-n_mJ_0(kb)J_0'(n_mkb)}{-J_0(n_mkb)H_0^{(1)}(kb)'+n_mH_0^{(1)}(kb)J_0'(n_mkb)}=ik^2\alpha_{zz}/4$$
$$\to\frac{-J_0(n_mkb)J_1(kb)+n_mJ_0(kb)J_1(n_mkb)}{J_0(n_mkb)Y_1(kb)-n_mY_0(kb)J_1(n_mkb)}=-k^2\alpha_{zz}/4\to$$
$$\frac{-(kb/2)+n_m^2(kb/2)}{-2/(\pi kb)}=-\pi(\varepsilon_m-1)(kb)^2/2=-k^2\alpha_{zz}/4\to$$
$$\alpha_{zz}=\pi b^2(\varepsilon_m-1)$$

If we include radiative loss (which amounts to going one order higher in the expansion of the Bessel functions of the first kind in the denominator), this result changes to:

$$\frac{(\varepsilon_m - 1)(kb/2)}{-2/(\pi kb) - i(kb/2) + i\varepsilon_m(kb/2)} = -k^2\alpha_{zz}/4 \rightarrow \frac{(\varepsilon_m - 1)\pi b^2}{1 - i(\varepsilon_m - 1)\pi(kb)^2/4} = \alpha_{zz}$$

This result is the same as **Equation 3** in the paper Extraordinary optical reflection from sub-wavelength cylinder arrays (https://opg.optica.org/directpdfaccess/87a6fce8-e489-4bee-bbbfeae2114ce881_89577/oe-14-9-3730.pdf?da=1&id=89577&seq=0&mobile=no).

Section 9: Generalized reflectance model

Although for general angles of incidence, it is more convenient to work with the magnetic field (since it is purely polarized along the cylinder axis while the electric field will have components along both the $\mathbf{e}_x$ and $\mathbf{e}_y$), for normal incidence it is just as convenient to work with the electric field (since it is purely polarized along the x axis). In this case, we take all the cylinders to be identical dipoles along the x axis. We write down a self consistency condition for each dipole moment (with E_0 denoting the incident electric field):

$$\mathbf{p}_n \cdot \mathbf{e}_x = \varepsilon_0\alpha \left[E_0 + (\omega^2\mu_0)\left[1 + \frac{c^2}{\omega^2}\partial_x^2\right]\sum_{m\neq n}\frac{i}{4}H_0^{(1)}(k|\mathbf{r}_n - \mathbf{r}_m|)\mathbf{p}_m \cdot \mathbf{e}_x\right]$$

We note that for two dimensional systems, α has units of length squared, which makes the above work out dimensionally. At this point, we may move all the terms proportional to the dipole moment to the left side of the equation to write out:

$$\mathbf{e}_x \cdot \mathbf{p}\left[1 - \varepsilon_0\alpha(\omega^2\mu_0)\left[1 + \frac{c^2}{\omega^2}\partial_x^2\right]\sum_{m\neq n}\frac{i}{4}H_0^{(1)}(k|\mathbf{r}_n - \mathbf{r}_m|)\right] = \varepsilon_0\alpha E_0$$

At this point, we use the following:

$$\left[1 + \frac{c^2}{\omega^2}\partial_x^2\right]\sum_{m\neq n}\frac{i}{4}H_0^{(1)}(k|\mathbf{r}_n - \mathbf{r}_m|) = -\frac{c^2}{\omega^2}\partial_y^2\sum_{m\neq n}\frac{i}{4}H_0^{(1)}(k|\mathbf{r}_n - \mathbf{r}_m|),$$

which simply follows from the fact that the scalar Green's function satisfies $\partial_x^2 G + (\omega/c)^2 G = -\partial_y^2 G$ (in two dimensions). This relation allows us to write the self consistency equation as:

$$\mathbf{e}_x \cdot \mathbf{p}\left[1 + \alpha\partial_y^2 \sum_{m\neq n} \frac{i}{4} H_0^{(1)}(k|\mathbf{r}_n - \mathbf{r}_m|)\right] = \varepsilon_0 \alpha E_0$$

$$\rightarrow \mathbf{e}_x \cdot \mathbf{p} = \frac{\varepsilon_0 \alpha E_0}{1 + \alpha\partial_y^2 \sum_{m\neq n} \frac{i}{4} H_0^{(1)}(k|\mathbf{r}_n - \mathbf{r}_m|)}$$

This allows us to define an effective polarizability in terms of the isolated scatterer's polarizability (similar to **Equations 24 and 25** in the paper: Extraordinary optical reflection from
sub-wavelength cylinder arrays (https://opg.optica.org/directpdfaccess/87a6fce8-e489-4bee-bbbfeae2114ce881_89577/oe-14-9-3730.pdf?da=1&id=89577&seq=0&mobile=no)):

$$\alpha_{eff} = \frac{\alpha}{1 + \alpha\partial_y^2 \sum_{m\neq n} \frac{i}{4} H_0^{(1)}(k|\mathbf{r}_n - \mathbf{r}_m|)} = \frac{1}{1/\alpha - \sum_{m\neq 0}(ik/4|\mathbf{r}_m|) H_1^{(1)}(k|\mathbf{r}_m|)}$$

Therefore, we may write the scattered field, $\mathbf{E}_{\text{scat}} \equiv E_{\text{scat}}\mathbf{e}_x$, as:

$$E_{\text{scat}}(z) = \frac{i(a\omega/c)}{2a^2} \frac{E_0 e^{-i|z|\omega/c}}{1/\alpha - \sum_{m\neq 0}(ik/4|\mathbf{r}_m|) H_1^{(1)}(k|\mathbf{r}_m|)}$$

From this, we may read off the reflection and transmission coefficients:

$$r(\omega) = \frac{i(a\omega/c)}{2a^2} \frac{1}{1/\alpha - \sum_{m\neq 0}(ik/4|\mathbf{r}_m|) H_1^{(1)}(k|\mathbf{r}_m|)}$$
$$t(\omega) = 1 + \frac{i(a\omega/c)}{2a^2} \frac{1}{1/\alpha - \sum_{m\neq 0}(ik/4|\mathbf{r}_m|) H_1^{(1)}(k|\mathbf{r}_m|)}$$

In order to get symmetric CPA, $r + t = 0$ (note that the antisymmetric CPA condition, $|r - t| = 0$ is obviously not possible), which gives us the following condition for the single particle polarizability, $\alpha(\omega)$:

$$-\frac{1}{\alpha(\omega)}+\sum_{m\neq 0}\frac{ik}{4a|m|}H_1^{(1)}(ka|m|)=\frac{i\omega}{ac}\rightarrow -\frac{i\omega}{ac}+\sum_{m\neq 0}\frac{ik}{4a|m|}H_1^{(1)}(ka|m|)=\frac{1}{\alpha(\omega)}$$

Since the polarizability has units of length squared, we write an expression in terms of the quantity $\tilde{\alpha}(\omega)\equiv\alpha(\omega)/a^2$, which is unitless:

$$\tilde{\alpha}(\omega)=\frac{-i(c/a\omega)}{\sum_{m>0}\frac{1}{2m}H_1^{(1)}(ma\omega/c)-1}=\frac{1}{\sum_{m>0}\frac{i(a\omega/c)}{2m}H_1^{(1)}(ma\omega/c)-i(a\omega/c)}$$

Therefore, we have the following for CPA:

$$1+\frac{i(ka)^2\tilde{\alpha}}{4}=\frac{n_mJ_1(n_mkb)H_1^{(2)}(kb)'-H_1^{(2)}(kb)J_1'(n_mkb)-\frac{i}{Z_sc\varepsilon_0}J_1'(n_mkb)H_1^{(2)}(kb)'}{H_1^{(1)}(kb)J_1'(n_mkb)-n_mH_1^{(1)}(kb)'J_1(n_mkb)+\frac{i}{Z_sc\varepsilon_0}J_1'(n_mkb)H_1^{(1)}(kb)'}$$

At this point, we solve for the conductivity:

$$n_mJ_1(n_mkb)\left[H_1^{(2)}(kb)'+sH_1^{(1)}(kb)'\right]-J_1(n_mkb)'\left[H_1^{(2)}(kb)+sH_1^{(1)}(kb)\right]+ \\ i(\sigma(\omega)/c\varepsilon_0)J_1'(n_mkb)\left[H_1^{(2)}(kb)'+sH_1^{(1)}(kb)'\right]=0$$

We specialize this for $n_m=1$:

$$J_1(kb)\left[H_1^{(2)}(kb)'+sH_1^{(1)}(kb)'\right]-J_1(kb)'\left[H_1^{(2)}(kb)+sH_1^{(1)}(kb)\right]+ \\ i(\sigma(\omega)/c\varepsilon_0)J_1'(kb)\left[H_1^{(2)}(kb)'+sH_1^{(1)}(kb)'\right]=0$$

We write this as $A+iB+i(\sigma(\omega)/c\varepsilon_0)(C+iD)$. We write this as:

$$A-C\tilde{\sigma}''(\omega)-D\tilde{\sigma}'(\omega)=0 \\ B+C\tilde{\sigma}'(\omega)-D\tilde{\sigma}''(\omega)=0$$

We write this in matrix form as:

$$\begin{bmatrix} -D & -C \\ C & -D \end{bmatrix} \begin{bmatrix} \tilde{\sigma}' \\ \tilde{\sigma}'' \end{bmatrix} = - \begin{bmatrix} A \\ B \end{bmatrix} \rightarrow \begin{bmatrix} \tilde{\sigma}' \\ \tilde{\sigma}'' \end{bmatrix} = - \frac{1}{D^2 + C^2} \begin{bmatrix} -D & C \\ -C & -D \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix}$$

Section 10: Sanity check (uncoupled scatterers)

In the absence of coupling, the reflection and transmission coefficients may be written entirely in terms of the single particle polarizabilities:

$$r(\omega) = \frac{i(a\omega/c)}{2a^2} \frac{1}{1/\alpha - \sum_{m\neq 0}(ik/4|\mathbf{r}_m|)H_1^{(1)}(k|\mathbf{r}_m|)} \rightarrow \frac{i(a\omega/c)\alpha}{2a^2}$$

$$t(\omega) = 1 + \frac{i(a\omega/c)}{2a^2} \frac{1}{1/\alpha - \sum_{m\neq 0}(ik/4|\mathbf{r}_m|)H_1^{(1)}(k|\mathbf{r}_m|)} \rightarrow 1 + \frac{i(a\omega/c)\alpha}{2a^2}$$

From the condition that $|r(\omega)|^2 + |t(\omega)|^2 < 1$, we obtain the following condition:

$$k^2|\alpha|^2/4a^2 + ik(\alpha - \alpha^*)/2a + k^2|\alpha|^2/4a^2 < 1 \rightarrow k^2|\alpha|^2 + ika(\alpha - \alpha^*) < 0$$

Since $\alpha - \alpha^* = 2i\Im\alpha$ and $\Im\alpha/|\alpha|^2 = -\Im(1/\alpha)$, we may conveniently write the above as:

$$-a\Im(1/\alpha) > k/2 \rightarrow a > (2/\pi)\lambda,$$

where we used the previously derived (in **Section 7**) expression for the polarizability,

$$\alpha = \frac{(\varepsilon_m - 1)(2\pi a^2)}{(\varepsilon_m + 1) - i(\varepsilon_m - 1)\pi(ka)^2/4} \rightarrow \Im(1/\alpha) = -k^2/8$$

And we also specialized to the case of no material loss.